

CHAPTER 7

HUMAN POPULATION AND THE ENVIRONMENT

7.1 POPULATION GROWTH

By United Nations count, the World's six billionth inhabitant was born in Sarajevo's Kosovo hospital at 12.01 a.m. on 12th October 1999. Although the precise date of birth of the six billionth world citizen cannot be determined, the United Nations has designated Tuesday as the Day of six billion in a consciousness raising exercise.

The growth of human population can be viewed in four major periods:

Stage 1: An early period of hunters and gatherers

This period ranges between the first evolution of humans on this planet to the beginning of agriculture. During this period, it is estimated that the total population was probably less than a million, population density was about 1 person per 130 – 260 sq km in the most habitable areas, and the average rate of growth was perhaps less than 0.00011% per year.

Stage 2: Early, pre-industrial agriculture

This period began some time between 9,000 BC and 6,000 BC and lasted approximately until the 16th century AD. The first major increase in population came during this period. The total human population of the world was approximately 100 million by 1 AD that further increased to about 500 million by 1600 AD. The population density increased greatly to about 1 or 2 people per sq km or even more. The average rate of growth was probably about 0.03%, which was large enough to increase the human population from approximately 5 million in 10,000 BC to about 100 million in 1 AD.

Stage 3: The age of industrial revolution

The second and much more rapid increase in population started about 400 years ago with the industrial revolution associated with the advances in medicines and health care. Experts are of the opinion that Renaissance in Europe, (beginning about 1600 AD) marks the transition from agricultural to literate societies, when medical care and sanitation were factors responsible in reducing the death rate. The total human population of about 900 million in 1800 AD, almost doubled in the next

century and again doubled to approximately 3 billion by 1960. The average rate of growth by 1600 AD was about 0.1% per year, which increased about one-tenth of a percent every 50 years until 1950. The main reasons of this rapid increase in population were discovery of the causes of diseases, invention of vaccines, improvement in sanitation, and advances in agriculture that led to a great increase in the production of food, shelter and clothing.

Stage 4 : The modern era

Though the rate of population growth has slowed down in developed countries, but population still continues to increase rapidly in many parts of the world, particularly developing countries. At mid 2002, the world population stood at 6.215 billions. The average rate of growth reached 2% in the middle of the 20th century and has declined to 1.3% in 2002 AD. According to projections, the global population will be approximately 8 billions by 2025 and 10 billions by 2050 AD.

According to World Bank projections, the world population may stabilize between 10.1 to 12.5 billions. Developed countries would only increase from 1.2 billions today to 1.9 billions, but developing countries would increase from 5 billions to 9.6 billions. In these projections, the developing countries will contribute 95% of the increase.

7.2 VARIATION AMONG NATIONS

In terms of continents and countries, the world's population is very ill-balanced. More than half of the world's people live in Asia (approximately 3.7 billion), which accounts for only one-fifth of the world's land area; while North, Central and South America together occupying more than a quarter of the land surface, have only one-fifth of the population (1.3 billion). The African continent also accounts for a quarter of the land surface but has just over one-eighth (840 million) of the world population. On the other hand, Europe whose area is only one twenty-fifth of the total has about one-ninth (728 million) of the world's people.

The distribution within the continents is also uneven. In Asia, China alone, with about 1.28 billion people, accounts for one-third Asian and one-fifth of the world population. The Indian subcontinent has a further 1.3 billion people—India, 1.05 billion; Pakistan, 143.5 million; Bangladesh, 133.6 million; Nepal, 23.9 million; Sri Lanka, 18.9 million; Bhutan, 0.9 million; and Maldives, 0.3 million. In Europe too, the population is unevenly distributed. Far less people live in Northern European countries than in other European countries. The most populous European countries are Russia (143.5 million), Germany (82.4 million), United Kingdom (60.2 million), France (59.5 million); Italy (58.1 million), Ukraine (48.2 million), Spain (41.3 million) and Poland (38.6 million). In

Africa and the Americas people are for the most part spread very thinly across the land, leaving large sections such as Northern Canada, southwest USA, the Sahara desert and the Amazon forest practically uninhabited.

The distribution of population depends to a large extent on the quality of land itself, which is very uneven. Thus population density (i.e., the number of people living in unit area) varies widely. The distribution of world population densities show that while the great majority of the land surface is sparsely or moderately populated (between 0 and 50 persons per sq km or between 0 and 125 ppsk), whereas some limited areas are densely populated. The densely populated areas include Western Europe, the Indian subcontinent, the plains and river valleys of China, and north-eastern USA. High concentrations of people are also found in some relatively smaller areas, for example, the Nile valley of Egypt, the Islands of Java in Indonesia and the Southern part of Japan.

The factors discouraging settlement are usually climatic or relief factors. The main factors are—cold, altitude, heat, drought, poor soils, inaccessibility, etc.

The factors encouraging settlements are—good land, flat or undulating terrain, the existence of mineral resources, a good climate suitable for a wide range of crops or a less equable climate suitable to the cultivation of specialized crops which have a good market, etc. Other factors include extension of roads, railways and other modes of transportation.

7.3 POPULATION EXPLOSION

The rapid growth of population is perhaps the most obvious factor affecting the present and future national and regional development, but it is by no means the only population problem in the world today. The main problem is that of *Population Explosion*.

Population explosion is a sudden spurt in the rate of population growth that occurs in the second stage of demographic transition due to a rapid fall in the mortality rate without a corresponding fall in the birth rate.

According to Theory of Demographic Transition, every country passes through three stages of demographic transition. In the first stage, both birth and death rates are high and hence population remains more or less stable. The main features of this stage are backward economy where agriculture is the main occupation, low per capita income, low standard of living, inadequate and unbalanced diet, absence of educational opportunities and presence of social evils like polygamy, early marriages etc. In the second stage, there is rapid growth of population. With the beginning of process of development, the living

standards of people improve, education expands, medical and health facilities increase as a result of which death rate comes down. But as long as society remains primarily agrarian and education remains confined to a narrow section of the society, attitude of the people towards the size of the family does not change radically and the birth rate remains high. In this situation, population increases at an alarming rate. If in a country the process of industrialisation accompanied by urbanisation is not fast and education is not widespread, then this is really a grave situation. This is called Population Explosion.

Population explosion doesn't mean overpopulation or population density. Overpopulation or population density is not the major problem. The problem arises when the economic development fails to maintain pace with population growth. So the most important factors regarding population are how fast population is growing; and most important is where it is growing. For example, Japan has a high population density but it ranked first on the human development index formulated by U.N.D.P. On the other hand low population density areas of Africa or S. America are unable to support the existing population. The most alarming finding is that the growth rate of these developing countries is much higher than the developed countries.

Thus, the size, distribution and structure of the population within a country must be viewed in relation to its natural resources and the techniques of production used by its population. The extent to which they are used and the way in which they are utilized determine whether an area/country is under- or over-populated and hence witnessing population explosion or not. A country is said to have an "optimum population" so long as the number of people is in balance with the available resources of the country.

7.3.1 Effects of Population Explosion

The effects of population explosion are numerous with far reaching consequences. Some of them are enumerated as under :

- unemployment,
- low living standard of people,
- hindrance in the process of development of economy,
- pressure on agriculture land,
- low per capita income,
- lack of basic amenities like water supply and sanitation, education, health, etc.,
- high crime rate
- environmental damage,
- migration to urban area in search of job,
- energy crisis,
- overcrowding of cities leads to development of slums.

7.3.2 Indian Scenario : A Case Study

Alarming growth of population is one of the most formidable problems India is facing today, and is seriously threatening its economic development. In 1921, the Indian population was 25 crores. It reached 84 crores in 1991 ; crossed the one billion mark with the billionth Indian born on August 15, 1999 ; and is expected to jump to 1.4 billion by 2030 AD. The population explosion, though a world wide phenomenon, poses a serious threat to India as it has to maintain 16.9% of world's population on only 2.4% of the world's area. The present growth rate of 1.7% is much higher than the world population growth rate of 1.3%, which is of great concern. The various reasons attributed to this great increase in population in India are :

- (i) Substantial drop in Death Rate due to advances in medical sciences. But the drop in Birth Rate is marginal. As a result of which the gap between death rate and birth rate widened.
- (ii) Universality of Marriage. In India marriage is universal to almost all men and women of marriageable age, as a result the birth rate in India is much higher.
- (iii) Due to practice of Early Marriage, there is a longer span for reproductivity.
- (iv) Due to Poverty, a poor man welcomes further additions to his family. He expects every member to earn and supplement the family income.
- (v) No Birth Control. Till recently, the use of contraceptives had been practically unknown in India.
- (vi) Existence of Polygamy.
- (vii) Tropical climate makes for early puberty.

The expected growth of population and its possible impact over next fifty years are summarized in Table 7.1.

In order to overcome this problem of population explosion, a sound Population Policy is required with the following objectives :

- (i) Quick economic development and raising the per capita income.
- (ii) Significant reduction in birth rate, which is more fundamental and important than the first, by providing legal and fiscal motivations like raising age of marriage, legalizing abortion etc.
- (iii) The planning of population must not aim merely at controlling the rate of multiplication but it should also include the improvement of the quality of the population as well by providing better facilities in education, health, etc.

(iv) The death rate should be brought down further, as high death rate results in waste of human energy and resources.

(v) Integrating population planning with economic planning.

We are thus facing a population explosion of crisis dimensions which has largely diluted the fruits of the remarkable economic progress that we have made over the last few decades. It is clear that simply to wait for education and economic development to bring about a desirable drop in fertility is not a practical solution. The time factor is so pressing and the population growth so formidable that we have to get out of this vicious circle through a direct assault upon the population problem as a national commitment.

Table 7.1. Growth of Population and its Impact Over Next 50 years

S. No.	Item	2011	2021	2031	2041	2051
1.	Projected Population (in Millions)	1178.4	1287.88	1394.66	1468.4	1516.86
2.	Demand of land for Residential use (in million ha.)	15.13	10.95	10.68	7.37	0.49
3.	Net land for Agriculture (in million ha.)	15.4	143.91	133.24	12.26	117.71
4.	Drinking water requirement (million litre/day)	159084	173863	188279	198234	204776
5.	Average Food Requirement per capita per day (million tonnes)	0.648	0.708	0.767	0.807	0.834
6.	No. of New Job seekers (in Millions)	730.60	811.36	892.58	954.46	1001.12
7.	Food Requirement (Million Tonnes per year)	236.56	258.54	279.97	394.78	304.50
8.	Food Production (Million Tonnes per year)	243.08	262.36	273.03	284.09	291.64
9.	Gain/Deficit Food Grains (Million Tonnes)	6.52	3.81	(-) 6.94	(-) 10.69	(-) 13.17